
CS 211P: Network and Database Practicum

Course Code: CS 211P

Course Name: Network and Database Practicum

L-T-P-C: 0-0-3-2

Category:

Prerequisites: None

Approval: 5th Senate

Course Contents

- Basic C Programming + Unix; notion of safety and importance of return values in function calls in C. Exposure to Unix command line and file system organization.
- Basic Unix System Calls.
- Unix sockets, ports, TCP/UDP end-to-end argument in networks; transport model in TCP and UDP; basic flow of server/client; TCP/UDP servers and clients.
- Basic Echo Server + File Server Design and implementation of a simple echo server using TCP and UDP; Chat program.
- Configuring a LAN, gateway sub netting; configuring multiple logical networks in the same physical network; basic commands like ping, ifconfig etc.
- Basics of Routing, IP Address Configuration, Subnetting - separation of logical networks using gateways; commands for configuring routing tables.
- Tcp dump, Wire shark; Observing network activity, tracing a connection; grouping connections by types/protocols/source/destination; writing filters; introduction to libpcap.
- Configuring and maintaining web server/DNS server.
- IPTables, Configuration, Setup notion of firewall; rule management; setting up and testing a simple firewall.
- Basics of Wireless Networks, setting up a real network, performance measurement.